

Effective resistivity of fully-processed SOI substrates

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Abstract

We introduce in this work a new quality factor called *effective resistivity* (ρ_{eff}), which is used to characterize and fairly compare the substrate resistivity of fully processed SOI wafers. The impacts on ρ_{eff} (and thus on microwave losses) of the bias (V_a), fixed oxide charges (Q_{ox}), traps at the SiO_2/Si interface (D_{it}), oxide thickness (t_{ox}) and line geometry are quantified and discussed for the first time. Different design and technological conclusions are drawn.

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1. Introduction

Keeping substrate losses at low levels is a priority target when designing high performance integrated silicon systems. In this field, high resistivity ($\rho_{\text{DC}} \geq 3 \text{ k}\Omega \text{ cm}$) silicon wafers are foreseen as promising candidates for low cost microwave applications [1,2]. However, oxide-passivated high resistivity (HR) wafers are known to suffer from parasitic surface conduction due to fixed charges (Q_{ox}) in the oxide [3]. The surface channel can also be formed underneath metallic lines with the application of a DC bias (V_a) [4]. An efficient way to reduce the concentration of free carriers near the Si surface is to introduce a large density of traps (D_{it}) at the SiO_2/Si interface. This has been successfully achieved by distinct methods, such as the modification of the oxide growth/deposition process parameters [5], the introduction of

an additional trap-rich layer between the oxide and the Si substrate (for instance, polysilicon in [6,7] or amorphous silicon in [8]), or also a selective etching of the oxide between the metallic conductors of coplanar waveguides (CPW) in [6,7]. As shown in Table 1, these substrate-engineering techniques lead to considerable RF loss reduction. However, assessing the efficiency of these technological solutions is not so trivial since in all of the above mentioned works total RF losses (α_{tot}) were reported, i.e. including conductor losses (α_{cond}), which become clearly dominant when dealing with high resistivity silicon substrates. Besides, conductor losses critically depend on metal thickness and conductivity, which as shown in the Table 1, significantly vary from one paper to the other. We therefore propose in this work a newly developed method to characterize the performance of the substrate with a factor of merit called *effective resistivity* (ρ_{eff}). The effects of Q_{ox} , D_{it} , V_a , oxide thickness (t_{ox}) and line geometry on ρ_{eff} are all investigated. Though the analysis is solely made on coplanar waveguides (CPW), it could easily be extended to other coplanar devices such as inductors or crosstalk structures.

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Table 1

Published experimental data of CPW losses (at 20 GHz) on HR substrates

Ref.	Substrate passivation layers	ρ_{DC} (k Ω cm)	Metal	Metal thickness (μ m)	W (μ m)	S (μ m)	α_{tot} (dB/cm)
[2]	None	2.5	Ti + Au	1.45	80	51	0.9
[6]	SiO ₂ (400 nm)	10	Al	1.3	70	40	17.2
[6]	Selectively etched SiO ₂ (400 nm)	10	Al	1.3	70	40	2.8
[6]	SiO ₂ (400 nm) + PolySi (600 nm)	10	Al	1.3	70	40	0.8
[6]	None	10	Al	1.3	70	40	1.9
[7]	Selectively etched SiO ₂ (100 nm)	10	Al	1	20	10	2.25
[7]	SiO ₂ (100 nm) + PolySi (300 nm)	10	Al	1	20	10	2.25
[8]	SiO ₂ (530 nm) + Ar implant (300 nm)	3	NA	4	50	25	1.9
[8]	SiO ₂ (530 nm) + α -Si (300 nm)	3	NA	4	50	25	1.3

2. Structure, model and effective resistivity extraction

The cross section and dimensions of the CPW structures considered in this work are given in Fig. 1a and b. In all cases, the passivation is made with silicon dioxide ($\epsilon_{r, SiO_2} = 3.9$) and the shunt distributed elements of the line consist in a classical combination of C_{ox_gnd} , C_{ox_sig} , C_{air} , C_{Si} and G_{Si} . For the case of very thick oxides, the transverse oxide capacitance becomes non negligible and must therefore be considered (C_{ox_tran}). In this work the structure is numerically studied and measured by biasing the line central conductor. The Si substrate surface is then locally altered by an accumulation, inversion and/or depletion layer. In order to fairly compare distinctly processed and biased Si wafers, we model the passivated and therefore *inhomogeneous* Si substrate with an effective *homogeneous* Si substrate (Fig. 1b) characterized by a uniform dielectric constant ($\epsilon_{r, Si} = 11.7$) and uniform resistivity, called *effective resistivity* (ρ_{eff}). The value of ρ_{eff} is such that the effective substrate has identical RF losses as the inhomogeneous, passivated wafer (i.e., $\alpha_{inh} = \alpha_{eff}$). Its extraction is based on the simplified model of the physical substrate presented in the inset of Fig. 1a ($C_{tot} || G_{tot}$).

For CPW lines made on passivated or naked Si substrates the quasi-TEM mode is dominant and RF substrate losses can accurately be estimated using respectively [9]:

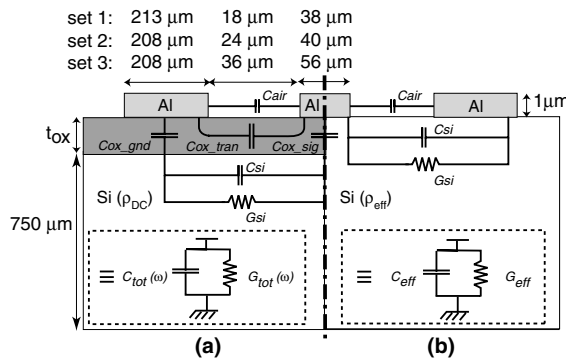


Fig. 1. Analyzed CPW structure on (a) a passivated Si substrate and (b) an effective homogeneous substrate used for ρ_{eff} modeling.

$$\alpha_{inh} \approx \frac{G_{tot}}{2} \sqrt{\frac{L}{C_{tot}}}; \quad \alpha_{eff} \approx \frac{G_{eff}}{2} \sqrt{\frac{L}{C_{eff}}} \quad (1)$$

where it is assumed that the lineic inductance (L) is identical for both structures. Since substrate losses are also considered to be equal, (1) yields:

$$G_{eff} = G_{tot} \sqrt{\frac{C_{eff}}{C_{tot}}} \quad (2)$$

where C_{eff} and G_{eff} are associated with the homogeneous substrate, and are therefore given by

$$C_{eff} = \epsilon_{r, eff} C_0 = (1 + q(\epsilon_{r, Si} - 1)) C_0 \quad (3)$$

$$G_{eff} = q \frac{C_0}{\epsilon_0 \rho_{eff}} = \left(\frac{\epsilon_{r, eff} - 1}{\epsilon_{r, Si} - 1} \right) \frac{C_0}{\epsilon_0 \rho_{eff}} \quad (4)$$

In these expressions q , $\epsilon_{r, eff}$ and C_0 are, respectively, the filling factor, the relative dielectric constant and the air-filled capacitance of the effective structure. Accurate close-forms analytical expressions that include metal thickness and finite ground plane width dependence are given by Heinrich in [10] for $\epsilon_{r, eff}$ and C_0 . Combining (2) and (4) leads to the following expression for ρ_{eff} :

$$\rho_{eff} = \frac{1}{\sqrt{\epsilon_{r, eff}}} \left(\frac{\epsilon_{r, eff} - 1}{\epsilon_{r, Si} - 1} \right) \frac{\sqrt{C_0}}{\epsilon_0} \frac{\sqrt{C_{tot}}}{G_{tot}} \quad (5)$$

in which G_{tot} and C_{tot} are extracted from measurements (or simulations) of the inhomogeneous studied structures.

3. Experimental results

The value of ρ_{eff} is extracted on distinctly processed SOI wafers (Table 2). All wafers are measured with an Anritsu VNA from 40 MHz to 40 GHz by biasing the central conductor of the CPW lines from -20 V to $+20$ V with 1 V steps (V_a). The propagation constant as well as the characteristic impedance of the lines are then extracted from the measured S -parameters with a classical TLR de-embedding technique [11] and an improved LL-method [12], respectively, enabling the com-

Table 2
Studied wafers

Wafer	t_{ox} (μm)	ρ_{DC} ($\Omega \text{ cm}$)	$\rho_{\text{eff}}^{\text{a}}$ ($\Omega \text{ cm}$)	Q_{ox} (cm^{-2})	D_{it} ($\text{eV}^{-1} \text{ cm}^{-2}$)	$\alpha_{\text{meas}}^{\text{a}}$ (dB/cm)
S1	0.15	>3k	~ 1000	$\sim 1 \times 10^{10}$	Low	2.71
S2	0.42	>3k	~ 1100	$\sim 1 \times 10^{10}$	Low	2.77
S3	0.42	>3k	~ 8000	$\parallel$	High	1.77
S4	0.15	>3k	~ 350	$\sim 2 \times 10^{10}$	Low	3.45
L1	2.7	>3k	~ 200	$\sim 20 \times 10^{10}$	Low	4.55
L2	2.7	~ 12	~ 22	$\sim 20 \times 10^{10}$	Low	17.8

^a Relates to 0 V and 20 GHz values, ρ_{DC} corresponds to the Si resistivity value extracted with the four-probe method.

putation of C_{tot} and G_{tot} . For each wafer ρ_{eff} is calculated with (5).

Since C_{tot} and G_{tot} are both frequency (f)-dependent, the effective resistivity varies with f as well. However, both C_{tot} and G_{tot} become frequency-independent when capacitive effects dominate inside the silicon substrate [13]. The effective resistivity therefore reaches a plateau level at high frequency as shown in Fig. 2. It is seen that this level is reached at higher frequencies for lower resistivity substrates, since the dielectric relaxation frequency is inversely proportional to the substrate resistivity. From now on the values of ρ_{eff} reported in the paper refer to the high frequency limit. Fig. 2 shows that the value of ρ_{eff} ($\sim 22 \Omega \text{ cm}$) measured on wafer L2 is somewhat higher than the nominal wafer resistivity ($\rho_{\text{DC}} = 12 \Omega \text{ cm}$ in Table 2), which is measured in DC at the back of the wafer by a four-probe method. This is simply due to the relatively thick oxide of the wafer. Fig. 2 also displays the values of ρ_{eff} extracted on wafer S4 (HR SOI wafer from which the Si film was removed by wet etching before metallization) for the three CPW sets considered in this work. The set dimensions are given in Fig. 1. It is seen that in all cases the value of ρ_{eff} is more than one order of magnitude lower than ρ_{DC} for that wafer (Table 2). As explained below, these low ρ_{eff} values are due to the presence of fixed charges in the oxide. However, Fig. 2 indicates that ρ_{eff} slightly in-

creases as the line dimensions are reduced. This effect is related to the higher C_0 value and the decrease of $\epsilon_{\text{r,eff}}$ for CPW lines with narrower spacings due to higher electric field concentration in the air region between the conductors. This also suggests that increasing ρ_{eff} could be further achieved by depositing thicker conductors on the oxide, which was not investigated here. Besides, for the geometries considered in this work we observe that the line dimensions do not significantly impact on ρ_{eff} and for this reason the rest of the analysis presented below will be performed on set 2 only.

Fig. 3 displays the values of ρ_{eff} as a function of the applied voltage for different wafers. In almost every cases we see that the ρ_{eff} versus V_{a} curve exhibits a peak value corresponding to the depletion of the silicon substrate underneath the line central conductor [5]. For higher or lower bias an accumulation or inversion layer is formed and ρ_{eff} is decreased. The S1 and S2 curves correspond to two identical HR UNIBOND SOI wafers from which the top Si film was respectively removed by wet etching and thermally oxidized in a wet ambient. In both cases a post metallization anneal (PMA) was performed in forming gas (10% H_2 , 90% N_2). We see that for both wafers the extracted value of ρ_{eff} is very close to $1 \text{ k}\Omega \text{ cm}$ despite a backside value higher than $3 \text{ k}\Omega \text{ cm}$, as measured with a four-probe method. The curves also indicate that only a minor improvement of ρ_{eff} is obtained on the thicker oxide structure. On the

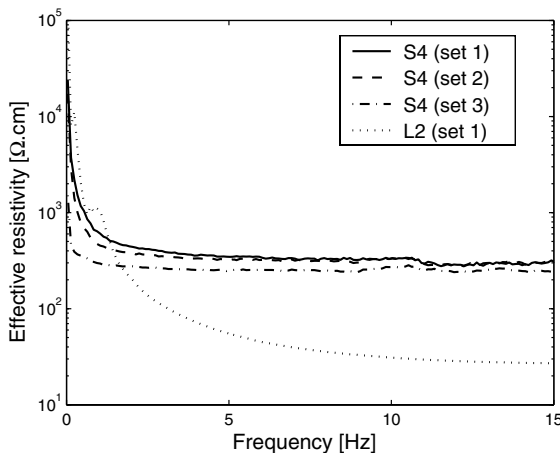


Fig. 2. Value of ρ_{eff} versus f extracted on wafers L2 and S4.

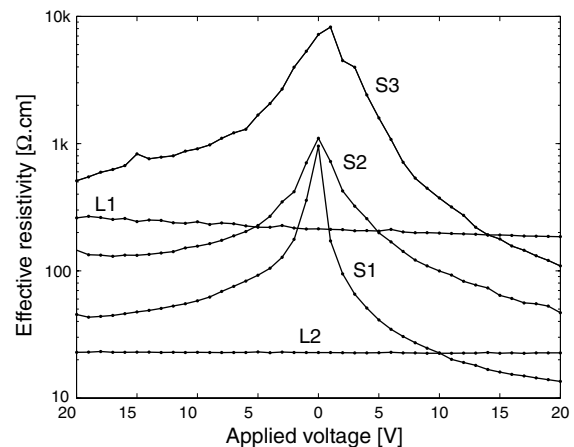


Fig. 3. Value of ρ_{eff} versus V_{a} extracted on the studied wafers.

other hand, a significant increase of ρ_{eff} is reached for the unannealed wafer. This is shown by the S3 curve recorded on a thermally oxidized SOI wafer without PMA, exhibiting ρ_{eff} values close to ρ_{DC} ($>3 \text{ k}\Omega \text{ cm}$). The effective resistivity is also extracted for a HR (L1) and standard (L2) SOI wafers where a full standard CMOS process with three metal levels was performed. On both wafers a $2.7 \mu\text{m}$ -thick oxide was deposited in several steps. We see that in both cases V_a has only a minor influence on the surface carrier concentrations, and thus on ρ_{eff} . As previously explained, the value of ρ_{eff} is slightly higher than ρ_{DC} for the standard resistivity wafer. A large resistivity decrease is however observed on wafer L1 for which ρ_{eff} remains close to $200 \Omega \text{ cm}$. This is more than one order of magnitude lower than ρ_{DC} for this wafer, despite an increased oxide thickness of $2.7 \mu\text{m}$. It is also worth noting that ρ_{DC} measured at the backside of the SOI wafers before and after the CMOS process remained unchanged. To bring additional light on those experimental data, Atlas [14] simulations were performed by investigating the impact of Q_{ox} , D_{it} and t_{ox} . The simulation results are presented next.

4. Simulations and discussion

(1) *Effects of Q_{ox}* : To quantify the effect of fixed oxide charges on the degradation of ρ_{eff} , the CPW of Fig. 1 (with $t_{\text{ox}} \simeq 150 \text{ nm}$ and $\rho_{\text{DC}} \simeq 3 \text{ k}\Omega \text{ cm}$) was first simulated with different values of Q_{ox} . The results are plotted in Fig. 4 as a function of V_a for an arbitrary frequency of 20 GHz . As expected all ρ_{eff} versus V_a curves are characterized by a sharp peak close to the flat band voltage (V_{FB}) of the structure. It is therefore shifted leftwards for increasing values of Q_{ox} . The figure highlights the enormous impact of oxide charges on ρ_{eff} : for Q_{ox} as low as $1 \times 10^{10} \text{ cm}^{-2}$ the maximum value of

ρ_{eff} is not higher than $800 \Omega \text{ cm}$. This is in close agreement with the extracted data from wafer S1 (Fig. 3).

(2) *Effects of D_{it}* : The simulations including interface traps were performed with a constant trap density throughout the Si forbidden bandgap. Donor traps were considered from the valence band edge up to a few tens of meV below midgap and acceptor traps in the rest of the bandgap. The results are presented in Fig. 5 for $\rho_{\text{DC}} \simeq 3 \text{ k}\Omega \text{ cm}$, $Q_{\text{ox}} = 1 \times 10^{10} \text{ cm}^{-2}$ and $t_{\text{ox}} = 150 \text{ nm}$. They show that for such value of Q_{ox} already significant improvements of ρ_{eff} are obtained with D_{it} as low as $1 \times 10^{10} \text{ eV}^{-1} \text{ cm}^{-2}$, because traps absorb free carriers from the interface and start depleting the Si surface. For higher D_{it} levels this effect is enforced and larger ρ_{eff} values are obtained. A stretch out of the peak is also observable in the ρ_{eff} versus V_a curves due to a reduced control of the surface potential by V_a when D_{it} is increased [15]. Stretch out and high ρ_{eff} values are clearly observed for wafer S3 (Fig. 3), suggesting a large trap density for this particular wafer where no PMA was performed. PMA is indeed known to passivate traps formed by Si dangling bonds at the SiO_2/Si interface with hydrogen atoms.

(3) *Effects of t_{ox}* : As shown in Fig. 6 increasing t_{ox} has a beneficial impact on ρ_{eff} (curves with + symbols) if the value of Q_{ox} is kept at constant value ($1 \times 10^{10} \text{ cm}^{-2}$ in this case), i.e., independently of t_{ox} . Unfortunately this does not occur in real structures made with current standard CMOS processes where several dielectric layers are stacked in order to increase the number of metal levels and isolate the upper metal from the semiconductor substrate. These layers are usually doped with boron or phosphorous to improve their mechanical properties [16], which is expected to yield distributed charges ($\rho(x)$) across the oxide and even larger defect densities. For a given charge distribution, an electrostatic analysis shows that the equivalent value of Q_{ox} is given by

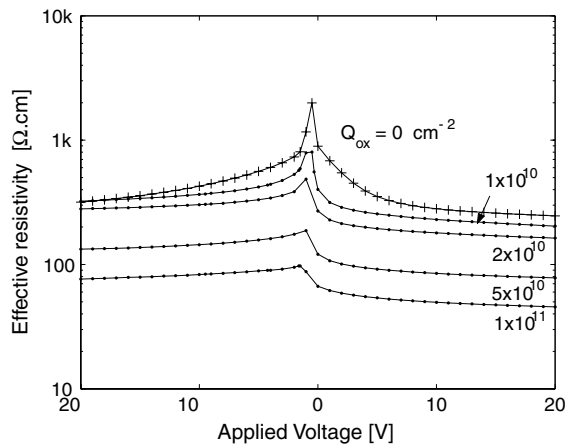


Fig. 4. Simulated values of ρ_{eff} as a function of V_a with $t_{\text{ox}} = 150 \text{ nm}$, varying Q_{ox} and $D_{\text{it}} = 0$.

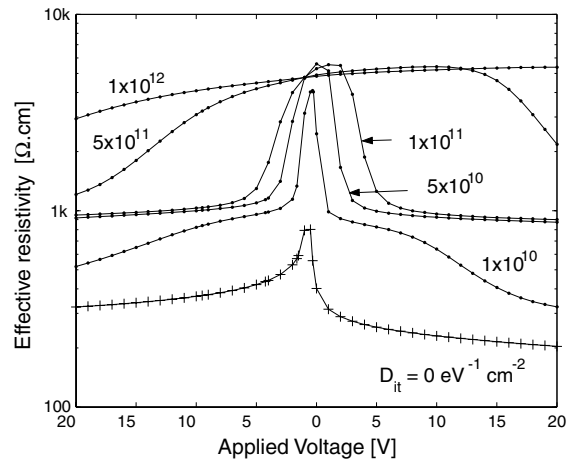


Fig. 5. Simulated values of ρ_{eff} as a function of V_a with $t_{\text{ox}} = 150 \text{ nm}$, $Q_{\text{ox}} = 1 \times 10^{10} \text{ cm}^{-2}$ and varying D_{it} .

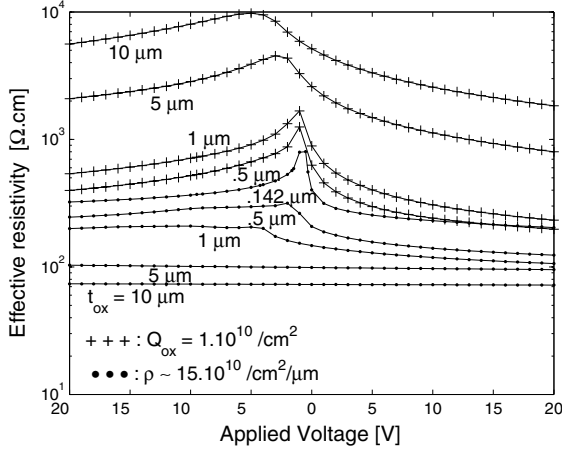


Fig. 6. Simulated values of ρ_{eff} as a function of V_a with varying t_{ox} , constant Q_{ox} or bulk fixed oxide charges and $D_{\text{it}} = 0$.

$$Q_{\text{ox}} = \int_0^{t_{\text{ox}}} \rho(x) \frac{x}{t_{\text{ox}}} dx \quad (6)$$

where the distance x is taken from the Al/SiO₂ interface. Assuming that the charges are equally distributed ($\rho(x) = \rho$) across the oxide, the level of Q_{ox} simply becomes $\rho t_{\text{ox}}/2$. This indicates that the resulting level of Q_{ox} linearly increases with t_{ox} .

Fig. 6 (curves with • symbols) illustrates the evolution of ρ_{eff} for a constant bulk oxide charge density of $\rho = 15 \times 10^{10} \text{ cm}^{-2} \mu\text{m}^{-1}$. In this case ρ_{eff} decreases with thicker oxides. For instance, a 5 μm -thick oxide has an equivalent Q_{ox} of $4 \times 10^{11} \text{ cm}^{-2}$ and exhibits a ρ_{eff} value as low as 100 $\Omega \text{ cm}$, which is more than one order of magnitude lower than ρ_{DC} . This simulation result is in total agreement with the large resistivity degradation observed on the fully processed L1 wafer. It therefore strongly suggests that this resistivity decrease is related to high Q_{ox} densities. Quasi-static C - V measurements have actually demonstrated that the equivalent level of Q_{ox} on L1 and L2 was as high as $2.8 \times 10^{11} \text{ cm}^{-2}$, leading to a bulk Q_{ox} close to $19 \times 10^{10} \text{ cm}^{-2} \mu\text{m}^{-1}$ and a flat band voltage of -35 V . This indicates that for HR SOI substrates parasitic surface conduction is even more critical when they are passivated with thick oxides ($\geq$ a few μm).

All these results demonstrate that a particular care must be taken when processing HR SOI substrates in order to prevent any degradation of the substrate resistivity. Yet, to this stage a question remains: what should be the level of ρ_{eff} in order to yield negligible losses inside the substrate? Fig. 7 provides a partial answer to the question by showing total CPW losses as a function of ρ_{eff} for different values of aluminum thickness. The figure also displays the level of conductor losses for each line thickness, assuming an aluminum conductivity of $3.5 \times 10^7 \text{ S/m}$. These data refer to set 2 and were computed at 20 GHz with analytical formulas given in

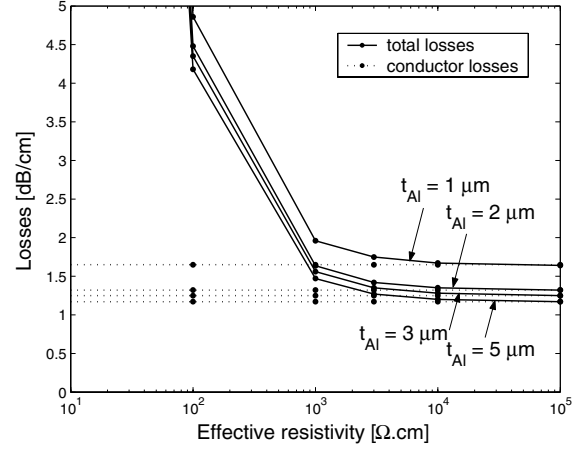


Fig. 7. Simulated total losses on set 2 as a function of ρ_{eff} for a varying value of aluminum thickness ($f = 20 \text{ GHz}$).

[10]. The results indicate that for an effective resistivity of 3 k $\Omega \text{ cm}$ substrate losses amount to less than 6% and 8% of α_{tot} for 1 and 5 μm metal thickness, respectively, making the substrate behave as quasi-lossless. This observation closely agrees with previously published data [1]. As expected also, these values decrease even further when ρ_{eff} increases and become clearly insignificant when ρ_{eff} reaches 10 k $\Omega \text{ cm}$. The figure also provides a quick view of the improvement obtained when the parasitic surface channel of the substrate can be suppressed. For instance a ρ_{eff} increase from 200 to 3000 $\Omega \text{ cm}$ leads to a loss reduction of around 58% in the case of 1 μm -thick lines, which is what could be expected on wafer L1. On the other hand, increasing the metal thickness from 1 to 5 μm only leads to a 17% loss reduction when ρ_{eff} is low (200 $\Omega \text{ cm}$). This shows that substrate losses still remain a major concern in current RFIC technologies, even in the case of HR bulk or SOI wafers. Substrates must therefore be provided with technological solutions to solve the issue of surface parasitic conduction and subsequent resistivity degradation. As shown in [6], this can be achieved by introducing a 300 nm-thick polysilicon layer between the oxide and the silicon. Such layer was indeed demonstrated to increase the value of ρ_{eff} up to 10 k $\Omega \text{ cm}$ [17].

5. Conclusion

We have introduced in this work a new quality factor called *effective resistivity* (ρ_{eff}). This quality factor was used to characterize and fairly compare the substrate resistivity of fully processed HR SOI wafers, leading to a better understanding of loss mechanisms occurring inside such substrates. Our results demonstrate for the first time that the effective resistivity of fully processed HR SOI wafers with several micron-thick oxides can be as low as 200 $\Omega \text{ cm}$. This value is more than one order

of magnitude lower than the wafer nominal resistivity ($\geq 3 \text{ k}\Omega \text{ cm}$) and leads to RF losses more than twice the level of RF losses that would be obtained on an $3 \text{ k}\Omega \text{ cm} - \rho_{\text{eff}}$ substrate. As shown by both experimental and simulation data, this degradation of substrate resistivity is closely associated with the large level of fixed charges present inside the poor quality interlayer dielectrics of current technologies. Indeed, these charges attract a low resistivity free carrier layer near the SiO_2/Si interface. Increasing the number of levels in stacked technologies to better isolate metallic conductors from the lines is therefore not an efficient solution, since it will lead to a further increase of Q_{ox} . Our simulation data therefore suggest that the most adequate technological solution to keep high values of ρ_{eff} ($\geq 3 \text{ k}\Omega \text{ cm}$) currently consists in the introduction of high trap densities at the SiO_2/Si ($\geq 5 \times 10^{11} \text{ eV}^{-1} \text{ cm}^{-2}$) interface in order to completely remove free carriers from the Si surface regardless of the oxidation type. To be effective, these traps should still remain active after the process is completed.

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